



Part 2: Network Security

Data Link Layer

DR. MUHAMMAD ISMAIL

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Functionalities

Data Link Layer Functionalities

- Overall, the data link layer implements three main functions:
 - (1) Framing Data: The OSI data link layer is intended to frame the data received from higher layers. (WHY?) Message frames are used to mitigate the data errors received from the physical layer.
 - (2) Node Addressing Support: In multi-node networks, the data link layer provides node addressing support so that nodes do not spend time processing message frames that are intended for other nodes.
 - (3) Transmission and Flow Control: The data link layer manages transmission and flow control for the physical layer to ensure that two nodes do not collide at the physical layer.
- *Remark: The data link layer functionalities are simplified for point-to-point networks. Ex of protocols include: Serial Line Internet Protocol (SLIP) and Point-to-Point Protocol (PPP).*

Data Flow (1/3)

- Transmitting Data:

- The data link layer receives data from the network layer (OSI layer 3) for transmission across the network.
- Because the node may generate data faster than the network transfer speed, the data link layer may cache data for transmission. But, if the transmission cache fills, then the data link layer informs the network layer that the data was not transmitted.
- To address potential data corruption, many data link protocols use an error detection scheme (e.g., checksum) and/or error correction system (e.g., Hamming code) for the data. These protocols encode the data and error mitigation information within the message frame. Framing Data
- The frame may also include addressing or routing information. Addressing Support

Data Flow (2/3)

- Transmitting Data:

- With most data link protocols, the message frames are uniform in length. This simplifies the receiving and processing of each frame. For high-speed networks, uniform length frames can result in a significant amount of padded network traffic. Frame relays resolve this issue by using variable-length frames.
- Finally, the data is passed from the data link layer to the physical layer for transmission across the physical network. If the network is temporarily unavailable due to network traffic or transmission delays, then the data link protocol will hold the data until the physical layer is ready to transmit.

Transmission and Flow Control

- Many protocols include a transmission timeout value. If the data cannot be transmitted in a reasonable amount of time, then the system may silently drop the packet or send an error code up the stack.

Data Flow (3/3)

- Receiving Data:

- The data is received from the physical layer.
- Each message frame is validated using the error detection (or correction) code. Invalid frames may generate a signal for a retransmission, or may prevent the transmission of a data link acknowledgement.
- If the frame contains address information, then the address is checked against the local node. Only information intended for the node is processed.
- Accepted messages (those with valid frames and appropriate addressing) have the frame removed. The data is placed into a receiving buffer for network layer.
- If the receiving buffer fills, then a flow control message is sent to the physical layer, or the receiving packet is lost. This situation may happen if the receiving node is not reading network data fast enough.

Point-to-Point and Multi-host Networks (1/2)

- Point-to-Point Network:

- Modems, microwave networks, or direct cable connections link two nodes directly.
- The data link message frame is used for error checking.
- For simplex physical mediums, only one node may transmit at a time, so the message frames ensure that one node does not dominate the network bandwidth.
- Duplex physical mediums permit both nodes to transmit at the same time. In this case, the message frame provides flow control, ensuring that a slow node does not drop packets when the receiving buffer fills.
- Examples of point-to-point data link protocols include the Point-to-Point Protocol (PPP) and Serial Link Internet Protocol (SLIP).

Point-to-Point and Multi-host Networks (2/2)

- Multi-host Networks:

- Any node connected to the physical layer may transmit data, and every node receives the transmitted data.
- In this environment, the message frame may include addressing formation. The addressing may indicate a single node recipient (unicast), multiple node recipients (multicast), or all nodes as recipients (broadcast). Most addressing schemes include the source (sender) and destination (recipients) in the message frame.
- The most common multi-host data link protocol is the media access control (MAC), defined by the IEEE 802.3 standard. The MAC provides a network-unique address (hardware address) used to identify a specific node on the physical network. To identify a node's hardware address, the data link layer includes an Address Resolution Protocol (ARP), as well as Reverse ARP (RARP).

Data Link Sublayers (1/4)

IEEE 802 splits the OSI data link layer into two sublayers: link layer control (LLC) and media access control (MAC).

- LLC Sublayer:

- It provides four functions: link management, service access point (SAP) management, connection management, and sequence management.
- Link Management: provides the control mechanisms for protocols that require flow control (e.g., Token Ring networks but not common for Ethernet). Also, Spanning Tree standard is used by bridges and ensures no routing loops.
- SAP: ports to network layer (OSI layer 3) protocols.
- Connection Management: determines whether the data link layer should use a connection-oriented or connection-less communication flow (indep. of TL).
- Sequence Management: ensures frames are in correct order.

Data Link Sublayers (2/4)

- MAC Sublayer:

- It provides two key functions: transmission control and network addressing.

- Transmission Control:

- Many of the different IEEE 802.x standards define the medium access control. For ex, IEEE 802.3 defines the Carrier Sense Multiple Access with Collision Detection (CSMA/CD) for Ethernet.
 - When a node has data to transmit, CSMA specifies that the node first listen to the network. If nobody else is transmitting, then the data is transmitted; however, CSMA provides no means to prevent two nodes from both hearing an empty network and transmitting at the same time. This results in a collision.
 - CD monitors the network for collisions. The node immediately stops transmitting, waits a few microseconds, and then repeats the process. A network error is passed to the higher layers if retransmission is not possible.

Data Link Sublayers (3/4)

- MAC Sublayer:

- Network Addressing:

- As each packet is received, the destination hardware (MAC) address is compared against the local hardware address. If the addresses match, then the packet is processed. Mismatches indicate a packet that is not intended for the node, so the node quietly drops the packet.
 - Ethernet network cards use a preset 6-byte code to specify the MAC address. The first three octets are the Organizationally Unique Identifier (OUI). These usually determine the manufacturer. The remaining three octets are vendor specific; they may indicate the particular make and model of the network card or a third-party vendor.

Data Link Sublayers (4/4)

- MAC Sublayer:
 - Network Addressing:
 - Some portion of the last three octets is pseudo-unique; two identical network cards purchased at the same time will likely have different 6- octet sequences. However, there are a limited number of unique octets.
 - To resolve potential address duplications, most network cards permit the network driver to override the preset MAC address. As such, any user can change the MAC address to any value.

Vulnerabilities and Attacks

Promiscuous Mode Attacks (1/3)

Data link layer attacks require direct physical access to the network. Because the data link layer does not extend beyond bridges, networks that are connected by routers or gateways are protected from these abuses. Remote attackers cannot directly attack the data link layer.

- Promiscuous Mode Attacks:

- Under normal usage, network-addressing schemes (e.g., MAC) prevent upper stack layers from receiving data that is not intended for the node.
- Many network interfaces support the disabling of the address filter. Nodes operating in promiscuous mode receive all message frames, not just those intended for the node → Promiscuous mode allows an attacker to receive all data from the network.
- Note: Promiscuous mode is commonly used as a network analysis and debugging tool by network administrators to view all local network traffic and identify malformed packets and routing issues.

Promiscuous Mode Attacks (2/3)

- Common attack vectors include the following:
 - Plaintext: Any data transmitted in plaintext (not encrypted) is readily viewable by an attacker.
 - Reconnaissance: An attacker may observe the quantity and types of systems on a network, types of traffic, and times when the network is active.
 - Hijacking: By knowing the type of data on a network, an attacker may attempt to take over an established connection.
 - Detection: By monitoring network traffic, attackers can gain an early warning that their presence has been detected.

Promiscuous Mode Attacks (3/3)

- Common Tools: Many tools that switch the network interface into promiscuous mode and capture network packets include:
 - Tcpdump: <http://www.tcpdump.org/>
 - Snort: <http://www.snort.org/>
 - Ngrep: <http://ngrep.sourceforge.net/> → scanning network traffic for specific byte sequences.
 - Wireshark/Ethereal: <http://www.ethereal.com/> → one of the most powerful tools that captures packets, reconstructs communications, and deciphers most protocols from OSI layers 2 and up.

Hardware Profiling Attacks

- The first step in attacking a system is reconnaissance. An attacker blindly attempting to attack an unknown system will rarely be successful.
- In a promiscuous mode, the attacker can collect information regarding the MAC addresses of victims. The OUI part of the MAC address provides hardware and operating system information to an attacker. For example, if an attacker sees a source MAC address with the OUI 00:0D:93, then the attacker knows the manufacturer is Apple Computers. The operating system associated with the MAC address is likely a version of MacOS and not Windows. Similarly, 00:20:F2 indicates Sun Microsystems—the host is likely running a version of SunOS or Solaris.
- When it is desirable to obscure the hardware profile, most network drivers permit changing the MAC address. The method to change the MAC address differs by operating system and may not be supported by some drivers

Address (Impersonation) Attacks

(1) Systems that use network addresses as access tokens can be quickly defeated. By observing the network in promiscuous mode, an attacker can identify the acceptable addresses. Because an attacker can change his MAC address, an attacker can readily assume any permitted MAC address. Similarly, an attacker may change his MAC address for the duration of an attack, which makes it difficult to track down the actual node. Authentication based on MAC addresses is weak at best. Address filtering should not be the only security option.

(2) An attacker may intentionally change the address to duplicate another node on the network. If both systems are active on the network, it is likely that both systems will interfere with each other. The attack effectively becomes a DoS.

Note: two nodes can coexist on the same network with the same hardware address. If both systems use different network layer (OSI layer 3) services or different data link SAPs, then they may operate without interference.

Load Attacks

- Attack appears as packets designed to increase a node's CPU load.
- In a multi-host network, every host receives every broadcast frame. These frames must be passed through the data link layer, where they are either managed by the upper data link layer protocols, or by higher OSI layers.
- The simple process of receiving and processing (or discarding) a single broadcast message frame does not consume many resources. Multiplying that same process by thousands of broadcast packets results in a significant overhead for the node.
- Attackers can quickly initiate a load attack because they can change their MAC address to be a broadcast or multicast address. All other nodes on the network will attempt to process every packet intended for the attacker's node.
- This attack is unlikely to be significant for regular workstations or bridges, however, it can be very serious for real-time or mission critical servers.

ARP Poisoning Attacks (1/4)

- ARP and RARP requests are broadcast packets that include a query. An ARP request contains an IP address. The expectation is that all nodes will receive the broadcast, but only one node will claim the IP address. The node that identifies the IP address will transmit an ARP reply (unicast) back to the querying host.
- Mapping IP addresses to hardware addresses (and vice versa) can add a delay to network connections. ARP tables contain a temporary cache of recently seen MAC addresses. Therefore, ARP packets are only required when a new IP address (or MAC address) lookup needs to be performed.
- ARP traffic is not passed beyond the local network. Bridges, routers, and gateways do not relay ARP packets. Nodes that are not on the local network will never be listed in the local ARP table.

ARP Poisoning Attacks (2/4)

- Unfortunately, these cached entries open the system to ARP poisoning, where an invalid or intentionally inaccurate entry is used to populate the table.
- ARP replies are not authenticated, but ARP reply values are placed into the ARP table. Moreover, if a later ARP reply is received, it can overwrite existing entries.
- The result from an ARP poisoning can range from a resource attack to a DoS or MitM attack.
 - Resource Attack:
 - Some systems have a limited number of ARP entries that can be cached. By sending a large number of false ARP entries, the ARP table can fill up. When the table fills, there are only two options: ignore additional ARP entries or throw out the oldest entries.

ARP Poisoning Attacks (3/4)

- Resource Attack:

- If the system ignores additional entries, then no new nodes on the local network can be contacted. If the node throws out the oldest ARP entry, then the result is a much slower network performance due to constant ARP queries for each packet that the system wants to transmit.

- Denial-of-Service (DoS) Attack:

- Newer ARP replies overwrite older ARP replies in the ARP table. If an entry in the ARP table is overwritten with a bad MAC address, then future connections to the overwritten node's IP address will fail.

ARP Poisoning Attacks (4/4)

- Man-in-the-Middle (MitM) Attack:
 - The MitM attack routes all traffic through a hostile node. Similar to the DoS attack, the ARP table entry is overwritten with a different machine's MAC address. In this situation, the new node will receive all traffic intended for the old node. By poisoning both nodes, the hostile node can establish a successful MitM.

Switch Attacks (1/2)

- Network devices such as switches and bridges commonly use ARP packets. In particular, these devices focus on ARP replies. Unfortunately, these systems are susceptible to ARP attacks such as *switch poisoning* and *switch flooding*.
- Switches maintain their own internal ARP table and associate each ARP reply with a physical port on the switch. Switches only retransmit the packet to every port when the destination MAC address is unknown.
- A switch can identify MAC addresses in three ways: (a) ARP Requests: Many switches know how to generate ARP requests. (b) ARP Replies: Even if not generated by the switch. (c) Frame Headers: As headers contain source MAC address, hence, associate source MAC address with specific switch port.

Switch Attacks (2/2)

- Switch Poisoning Attack:

- Switches maintain an ARP table to route traffic through the switch to specific physical network ports. An ARP poisoning attack can corrupt the ARP table. The poisoning ARP reply can associate another node's MAC address with a different port. This effectively cuts off the victim node from the network and sends all traffic intended for the victim through the attacker's port. Because switch poisoning redirects network traffic, this attack allows connection hijacking.

- Switch Flooding Attack:

- An attacker, using ARP poisoning, can flood a switch's ARP table. Because switches cannot afford to drop packets, most switches regress to a hub state when the switch's ARP table fills. Hence, all nodes receive all traffic; a node operating in promiscuous mode can receive all network traffic that passes through the switch.

Covert Channels (1/2)

- A covert channel is a type of computer attack that allows the communication of information through not normally used or unintended ways. The goal is to convey information while being undetected or to bypass security policies. Exs include:
 - Out-of-Frame-Data:
 - In general, data that is not enclosed in a message frame is discarded. Information can be transmitted across the physical layer and not be included in a message frame. This out-of-frame data can consume network bandwidth or convey information in a nonstandard (covert) fashion.
 - Network interfaces operating in promiscuous mode may receive the out-of-frame data

Covert Channels (2/2)

- Invalid Frames:
 - Invalid frames are normally discarded. A covert channel may intentionally create frames with invalid checksums.
- Frame Size:
 - For networks that support variable frame sizes, the size or padding within a frame may be used to pass subtle information that is difficult to detect.
- Remember: Data link covert channels are usually restricted to the local network.

Mitigation Techniques

Data Link Authentication (1/2)

- Few data link layer protocols implement cryptography; the processing overhead for cryptographic data link protocols can be very expensive and inadvertently result in a load attack for busy networks.
- Instead, the two main cryptographic solutions are CHAP and PAP, which are primarily used for point-to-point connections.
 - CHAP:
 - The Challenge-Handshake Authentication Protocol (CHAP) uses a shared key to authenticate the initial network connection: (1) The client connects to the server. (2) The server issues a challenge. This is a value based on a one-way hash using the secret key. (3) The client validates the challenge by comparing it with the expected hash value. (4) The client sends its own hash to the server. (5) The server validates the response.

Data Link Authentication (2/2)

- CHAP:

- Only fully validated connections are permitted. CHAP provides a means to authenticate two nodes, but it does not perform any validation or encryption after the connection is established.

- PAP:

- The Password Authentication Protocol (PAP) is a much simpler than CHAP.
- The login credentials (id and password) are transmitted in plaintext from the client to the server. If the server accepts the credentials, then the connection is permitted.

Promiscuous Mode Detection (1/2)

- Promiscuous mode does have one significant limitation in that many higher layer protocols assume that the lower layers have vetted the network data. If the data within a packet requires a reply, higher layer protocols may reply.

- Detection Using ARP:

- ARP packets are used to associate IP addresses (OSI layer 3) with hardware addresses (OSI layer 2). Under normal usage, the receiving node first checks the hardware address. If the address indicates that it is intended for the node, then the ARP's IP address query is processed. If the hardware address does not match the host, then the data link layer normally rejects the packet. In promiscuous mode, however, the packet is accepted and an ARP response is generated.

Promiscuous Mode Detection (2/2)

- Detection Using ICMP:

- The ICMP Echo packet (also known as a ping packet) normally generates an ICMP reply from an active host.
- Multiple ICMP packets should generate multiple replies with similar latencies.
- Under promiscuous mode, the node receives more packets to process and therefore has a larger latency.

- Detection Using DNS:

- Many packet-capturing tools perform hostname lookups for IP addresses that are captured. Monitoring DNS requests can determine which hosts are performing many hostname lookups. This may indicate a host that is capturing packets.

Mitigating ARP Poisoning (1/2)

- Hard-Coding ARP Tables:

- Normally ARP tables are populated dynamically, as each ARP reply is received; however, ARP tables can be populated statically using the operating system's arp command. This command statically sets a MAC address and IP address pair into the ARP table and prevents future modification.

- Expiring ARP Entries:

- Cached entries in an ARP table may timeout. Expired entries, where the same MAC and IP address combination are not observed, can be removed from the ARP table. This mitigation option limits the impact from resource attacks.

Mitigating ARP Poisoning (2/2)

- Filtering ARP Replies:

- Not every ARP reply must be inserted into the ARP table. Linux and BSD limit cache entries to ARP requests that were sent by the local host. This prevents unsolicited ARP replies from entering the ARP table. In contrast, most Windows operating systems accept unsolicited ARP replies and insert them into the table.
- Although ARP reply filtering does prevent unsolicited entries, it does not prevent new replies from overwriting existing entries.

- Locking ARP Tables:

- ARP tables can be temporarily locked. In this situation, an established connection (such as an IP connection) locks the ARP table entry for a short duration. During this time, new ARP replies cannot overwrite the locked table entry.

Mitigating Switch Attacks

- Most bridges and high-end switches support static ARP entries. Using static ARP tables in these network devices can mitigate the impact from switch flooding and poisoning.
- Although flooding attacks can enable network monitoring by a promiscuous node, segmenting the network with multiple switches and bridges still mitigates the impact from this attack. For example, an onion network topology can have any single bridge/switch overwhelmed by a flooding attack, but not everyone will be compromised. As each device regresses to a hub state, the overall network volume increases. Eventually the network will become unusable due to too much network traffic; the onion topology will limit the DoS to a few rings. In addition, IDS and IPS monitors will likely detect a sudden increase in network traffic (and ARP replies), indicating a network attack.

THANK YOU!